

# Supplementary Information

Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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All cases, scripts and tabulated results: [github.com/Imerson/cryostat-first-shield-He-vs-LN2](https://github.com/Imerson/cryostat-first-shield-He-vs-LN2).

## S1 Enclosure model

**Table S1.** Regions, patches and boundary conditions of the enclosure CHT cases.

| Region                   | Patch                    | $T$ condition                              | Radiation condition                |
|--------------------------|--------------------------|--------------------------------------------|------------------------------------|
| domain0 (OVC-shield gap) | warmPlate, OVC           | fixed $T_{\text{hot}}$                     | grey diffuse, $\varepsilon = 0.04$ |
|                          | shield_slave             | fixed $T_{\text{cold}}$                    | grey diffuse, $\varepsilon = 0.02$ |
| domain1 (inner gap)      | warmPlate                | fixed $T_{\text{hot}}$                     | grey diffuse, $\varepsilon = 0.04$ |
|                          | coldPlate, shield_bottom | fixed $T_{\text{cold}}$                    | grey diffuse, $\varepsilon = 0.04$ |
|                          | shield                   | fixed $T_{\text{cold}}$                    | grey diffuse, $\varepsilon = 0.02$ |
|                          | domain1_to_coax_Lk       | conjugate (mapped)                         | grey diffuse, $\varepsilon = 0.02$ |
| coax_L1..L4 (solids)     | warm cap / cold cap      | fixed $T_{\text{hot}}$ / $T_{\text{cold}}$ | –                                  |
|                          | lateral                  | conjugate to domain1                       | –                                  |

**Table S2.** Gas-region settings that make the gaps behave as vacuum. Radiation in OpenFOAM’s `viewFactor` model is solved in fluid regions, so each gap is a fluid region made inert.

| Setting                       | Value                                                                                            |
|-------------------------------|--------------------------------------------------------------------------------------------------|
| pressure $p = p_{\text{rgh}}$ | $10^{-3}$ Pa ( <code>pRefValue</code> $10^{-3}$ )                                                |
| gravity                       | (0, 0, 0)                                                                                        |
| transport                     | constant, $\mu = 10^{-11}$ Pa s, $\text{Pr} = 0.67$                                              |
| density limits                | $\rho_{\text{max}} = 10^{-4}$ , $\rho_{\text{min}} = 10^{-12}$ kg m $^{-3}$                      |
| radiation solver frequency    | every 10 iterations                                                                              |
| agglomeration                 | <code>nFacesInCoarsestLevel</code> 250, <code>featureAngle</code> 20, <code>mergeLevels</code> 1 |

**Table S3.** Harness representation. Cable UT-085-SS-SS: 304 stainless outer conductor  $A = 1.585 \text{ mm}^2$ , PTFE dielectric  $A = 2.001 \text{ mm}^2$ , 304 stainless centre conductor  $A = 0.205 \text{ mm}^2$ . Material  $k(T)$  from the NIST cryogenic fits  $\log_{10} k = \sum_n a_n (\log_{10} T)^n$ . Each modelled cylinder carries  $n_k$  lines with  $\kappa_{\text{eff}}(T) = (n_k/A_{\text{model}}) \sum_i A_i k_i(T)$ .  $A_{\text{model}}$  is the conduction cross-section of the meshed cylinder in each stage span (calibrated from the mesh so that  $(A_{\text{model}}/L) \int \kappa_{\text{eff}} dT$  equals the physical cable integral  $n_k L^{-1} \sum_i A_i \int k_i dT$ ); it differs between the stage-1 and stage-2 geometries. Because  $A_{\text{model}}$  cancels in that product, the stage-integrated conduction is fixed by the cable data by construction: the resolved model reproduces the targets below to 0.01 %, which verifies the implementation but is not an independent prediction.

| Span     | $A_{\text{model}} [\text{m}^2]$ | $L [\text{m}]$ | L1 drive, $n_k=25$ | L2 flux, 25 | L3 readout, 10 | L4 pump, 5 |
|----------|---------------------------------|----------------|--------------------|-------------|----------------|------------|
| 300→50 K | $2.975 \times 10^{-4}$          | 0.335          | 395.8 mW           | 395.8 mW    | 158.3 mW       | 79.2 mW    |
| 300→77 K | $2.976 \times 10^{-4}$          | 0.335          | 370.0 mW           | 370.0 mW    | 148.0 mW       | 74.0 mW    |
| 50→4 K   | $3.934 \times 10^{-3}$          | 0.245          | 27.03 mW           | 27.03 mW    | 10.81 mW       | 5.41 mW    |
| 77→4 K   | $2.845 \times 10^{-3}$          | 0.245          | 62.33 mW           | 62.33 mW    | 24.93 mW       | 12.47 mW   |

Per-bundle targets  $n_k L^{-1} \sum_i A_i \int k_i dT$ ; shares 38.5 / 38.5 / 15.4 / 7.7 % follow from  $n_k$  with identical cable construction and length.

**Table S4.** Deployed  $\kappa_{\text{eff}}(T) = \sum_{n=0}^5 c_n T^n [\text{W m}^{-1} \text{K}^{-1}]$ ,  $T$  in K] per bundle and stage span (`kappaCoeffs<8>`, last two coefficients zero). Fit error < 0.5 % over each span.

| Span     | Bundle | $c_0$                    | $c_1$                   | $c_2$                    | $c_3$                    | $c_4$                     | $c_5$                     |
|----------|--------|--------------------------|-------------------------|--------------------------|--------------------------|---------------------------|---------------------------|
| 300→50 K | L1     | $-2.1522 \times 10^{-1}$ | $3.1627 \times 10^{-2}$ | $-2.3184 \times 10^{-4}$ | $9.9921 \times 10^{-7}$  | $-2.1908 \times 10^{-9}$  | $1.9383 \times 10^{-12}$  |
|          | L2     | $-2.1519 \times 10^{-1}$ | $3.1622 \times 10^{-2}$ | $-2.3181 \times 10^{-4}$ | $9.9907 \times 10^{-7}$  | $-2.1905 \times 10^{-9}$  | $1.9380 \times 10^{-12}$  |
|          | L3     | $-8.6092 \times 10^{-2}$ | $1.2651 \times 10^{-2}$ | $-9.2741 \times 10^{-5}$ | $3.9970 \times 10^{-7}$  | $-8.7636 \times 10^{-10}$ | $7.7535 \times 10^{-13}$  |
|          | L4     | $-4.3044 \times 10^{-2}$ | $6.3253 \times 10^{-3}$ | $-4.6369 \times 10^{-5}$ | $1.9984 \times 10^{-7}$  | $-4.3816 \times 10^{-10}$ | $3.8766 \times 10^{-13}$  |
| 300→77 K | L1     | $-1.3459 \times 10^{-1}$ | $2.9103 \times 10^{-2}$ | $-2.0205 \times 10^{-4}$ | $8.3171 \times 10^{-7}$  | $-1.7398 \times 10^{-9}$  | $1.4706 \times 10^{-12}$  |
|          | L2     | $-1.3457 \times 10^{-1}$ | $2.9099 \times 10^{-2}$ | $-2.0202 \times 10^{-4}$ | $8.3159 \times 10^{-7}$  | $-1.7396 \times 10^{-9}$  | $1.4704 \times 10^{-12}$  |
|          | L3     | $-5.3837 \times 10^{-2}$ | $1.1642 \times 10^{-2}$ | $-8.0824 \times 10^{-5}$ | $3.3269 \times 10^{-7}$  | $-6.9596 \times 10^{-10}$ | $5.8824 \times 10^{-13}$  |
|          | L4     | $-2.6917 \times 10^{-2}$ | $5.8206 \times 10^{-3}$ | $-4.0410 \times 10^{-5}$ | $1.6634 \times 10^{-7}$  | $-3.4796 \times 10^{-10}$ | $2.9411 \times 10^{-13}$  |
| 50→4 K   | L1     | $-4.8624 \times 10^{-4}$ | $9.1378 \times 10^{-4}$ | $3.6621 \times 10^{-5}$  | $-9.0343 \times 10^{-7}$ | $8.4798 \times 10^{-9}$   | $-2.8699 \times 10^{-11}$ |
|          | L2     | $-4.8624 \times 10^{-4}$ | $9.1378 \times 10^{-4}$ | $3.6621 \times 10^{-5}$  | $-9.0343 \times 10^{-7}$ | $8.4798 \times 10^{-9}$   | $-2.8699 \times 10^{-11}$ |
|          | L3     | $-1.9450 \times 10^{-4}$ | $3.6551 \times 10^{-4}$ | $1.4648 \times 10^{-5}$  | $-3.6137 \times 10^{-7}$ | $3.3919 \times 10^{-9}$   | $-1.1480 \times 10^{-11}$ |
|          | L4     | $-9.7247 \times 10^{-5}$ | $1.8275 \times 10^{-4}$ | $7.3242 \times 10^{-6}$  | $-1.8068 \times 10^{-7}$ | $1.6959 \times 10^{-9}$   | $-5.7398 \times 10^{-12}$ |
| 77→4 K   | L1     | $-7.4665 \times 10^{-4}$ | $1.2814 \times 10^{-3}$ | $4.9472 \times 10^{-5}$  | $-1.2245 \times 10^{-6}$ | $1.1714 \times 10^{-8}$   | $-4.2617 \times 10^{-11}$ |
|          | L2     | $-7.4665 \times 10^{-4}$ | $1.2814 \times 10^{-3}$ | $4.9472 \times 10^{-5}$  | $-1.2245 \times 10^{-6}$ | $1.1714 \times 10^{-8}$   | $-4.2617 \times 10^{-11}$ |
|          | L3     | $-2.9865 \times 10^{-4}$ | $5.1254 \times 10^{-4}$ | $1.9788 \times 10^{-5}$  | $-4.8978 \times 10^{-7}$ | $4.6855 \times 10^{-9}$   | $-1.7046 \times 10^{-11}$ |
|          | L4     | $-1.4932 \times 10^{-4}$ | $2.5626 \times 10^{-4}$ | $9.8938 \times 10^{-6}$  | $-2.4488 \times 10^{-7}$ | $2.3427 \times 10^{-9}$   | $-8.5230 \times 10^{-12}$ |

**Free-molecular gas conduction bound.** At  $p < 10^{-2} \text{ Pa}$  the mean free path exceeds the gap and gas heat transport is free-molecular,  $\dot{Q}_{\text{gas}} = \alpha \Lambda p A \Delta T$  [1], with  $\Lambda = \frac{\gamma+1}{\gamma-1} \sqrt{R/(8\pi MT)}$  evaluated at the warm wall (helium,  $\gamma = 5/3$ :  $\Lambda = 2.10 \text{ W m}^{-2} \text{K}^{-1} \text{Pa}^{-1}$ ; nitrogen: 1.19), accommodation coefficient  $\alpha$ , exchange area  $A$  and  $\Delta T = 250 \text{ K}$ . For the enclosure's equivalent plate area  $A = 1.14 \text{ m}^2$  and helium with  $\alpha = 0.4$  the bound is 2.4 mW at  $10^{-5} \text{ Pa}$ , 24 mW at  $10^{-4} \text{ Pa}$  and 0.24 W at  $10^{-3} \text{ Pa}$ , that is 0.2, 2.3 and 23 % of the 1.03 W harness conduction ( $\alpha = 1$ : 0.6, 5.8 and 58 %). The model therefore applies to a cryostat pumped to  $10^{-4} \text{ Pa}$  or better, where the residual gas is a budget line of a few per cent; the nominal  $p = 10^{-3} \text{ Pa}$  of the gap regions is a solver setting for the radiation-only vacuum, not an operating pressure.

**Table S5.** View-factor agglomeration refinement and row normalisation. Closure = mean row sum  $\sum_j F_{ij}$  of the generated matrix (unity for a closed enclosure); loads with the matrix as generated and, at the production level, with rows normalised.

| Case     | Coarsest-level faces | Closure (mean; range) | Radiation, as generated |                            | Radiation, normalised |
|----------|----------------------|-----------------------|-------------------------|----------------------------|-----------------------|
| 300→50 K | 60                   | –                     | 9.46 W                  | –                          |                       |
| 300→50 K | 250 (production)     | 0.99–1.01; 0.50–1.13  | 8.41 W                  | 7.36 W                     |                       |
| 300→50 K | 400                  | 0.98–0.99             | 6.49 W                  | 7.32 W                     |                       |
| 300→50 K | 800                  | 0.90–0.96             | 3.26 W                  | 7.27 W                     |                       |
| 300→50 K | closed form          | –                     | 7.07 W                  |                            |                       |
| 300→77 K | 250 (production)     | 0.99                  | 8.38 W                  | 7.34 W (closed form 7.04)  |                       |
| 50→4 K   | 250 (production)     | 1.00                  | 5.29 mW                 | 3.16 mW (closed form 3.15) |                       |
| 77→4 K   | 250 (production)     | 1.00                  | 29.7 mW                 | 17.8 mW (closed form 17.7) |                       |

**Conservation and reciprocity audit.** Two residuals are reported for every view-factor matrix. Global conservation: the net radiative power  $\sum_{\text{patches}} \int q_r dA$  summed over all surfaces of a closed region, which must vanish; the generated matrices leave 1.5 W (400 faces) and 14 W (800 faces) unbalanced on the 300→50 K case, the row-normalised ones 0.10, 0.13 and 0.36 W at 250, 400 and 800 faces. Reciprocity: with the coarse-face areas  $A_i$  rebuilt from the mesh by summing the fine faces of each agglomeration (`scripts/reciprocity.py`), the area-weighted error  $\sum |A_i F_{ij} - A_j F_{ji}| / \sum A_i F_{ij}$  of the normalised matrices is 3.9, 5.5 and 12.6 % at 250, 400 and 800 faces (outer gap region). Row normalisation restores closure exactly but leaves reciprocity untouched, and both residuals grow with agglomeration refinement, which is why the production level is the coarsest, 250 faces, and why the 3–4 % departure from the closed form is quoted as the accuracy of the method.

**Row normalisation.** Each fluid region of the enclosure model is a closed surface, so every row of the view-factor matrix must sum to unity. The generated matrix (`viewFactorsGen`, Gauss quadrature tolerance 0.01) is stored sparsely, one entry per visible face pair; the normalisation divides each row by its sum and leaves rows with a sum below 0.2 (none occurred) untouched. The solve was then restarted from the converged production field for 600 further iterations. Script `normalise_F.py` in the repository.

**Table S6.** Emissivity sensitivity (production view factors) and MLI-equivalent variant; the MLI-equivalent cases are also given with the row-normalised matrix, which is the value carried into the main text and the ledger.

| Case                               | $\varepsilon$ warm / cold | Radiation [W] | Total [W] |
|------------------------------------|---------------------------|---------------|-----------|
| 300→50 K, low                      | 0.02 / 0.01               | 5.45          | 6.48      |
| 300→50 K, baseline                 | 0.04 / 0.02               | 8.41          | 9.44      |
| 300→50 K, high                     | 0.06 / 0.03               | 11.48         | 12.51     |
| 300→77 K, low                      | 0.02 / 0.01               | 5.44          | 6.40      |
| 300→77 K, baseline                 | 0.04 / 0.02               | 8.38          | 9.35      |
| 300→77 K, high                     | 0.06 / 0.03               | 11.44         | 12.40     |
| 300→50 K, MLI-eq.                  | 0.04 / 0.003              | 1.81          | 2.84      |
| 300→77 K, MLI-eq.                  | 0.04 / 0.003              | 1.81          | 2.77      |
| 300→50 K, MLI-eq., rows normalised | 0.04 / 0.003              | 1.69          | 2.72      |
| 300→77 K, MLI-eq., rows normalised | 0.04 / 0.003              | 1.68          | 2.65      |

## S2 Cold-plate sweep: full results

**Table S7.** All cold-plate runs at 9.4 W (production mesh unless marked).  $\dot{m}$  in  $\text{g s}^{-1}$ , volumetric flow in  $\text{cm}^3 \text{s}^{-1}$ ,  $\Delta p$  in Pa,  $\dot{W}_p$  in  $\mu\text{W}$ , temperatures in K,  $h$  in  $\text{W m}^{-2}\text{K}^{-1}$ ;  $T_{\text{in}} = 45 \text{ K}$  (He),  $77 \text{ K}$  ( $\text{LN}_2$ ).  $T_w$  is the area mean over the heated-wall patch,  $T_{w,\text{max}}$  and  $T_{w,\text{min}}$  its extreme face values (`surfaceFieldValue` on the patch);  $\text{margin} = T_{\text{sat}}(5 \text{ bar}) - T_{w,\text{max}} = 94.0 \text{ K} - T_{w,\text{max}}$ ;  $y^+ =$  maximum on the heated wall; closure = energy-balance error. Gravity-on (g on) nitrogen runs did not reach a steady state (see closure).

| Case                                        | $Re$  | $\dot{m}$ | $\dot{V}$ | $\Delta p$ | $\dot{W}_p$ | $T_{\text{out}}$ | $T_w$ | $T_{w,\text{max}}$ | $T_{w,\text{min}}$ | $T_w - T_b$ | $h$  | $Nu$ | $f$    | NTU    | $\varepsilon$ | Gr/ $Re^2$ | margin | $y^+$ | closure % |
|---------------------------------------------|-------|-----------|-----------|------------|-------------|------------------|-------|--------------------|--------------------|-------------|------|------|--------|--------|---------------|------------|--------|-------|-----------|
| He 1 bar $Re=500$                           | 500   | 0.159     | 167       | 0.294      | 48.6        | 56.36            | 87.4  | 106.1              | 47.9               | 36.72       | 17.1 | 3.66 | 0.186  | 0.31   | 0.266         | 0.63       | —      | —     | -0.06     |
| He 1 bar $Re=500$ (g on)                    | 500   | 0.159     | 167       | 0.331      | 54.6        | 56.21            | 67.5  | 95.2               | 47.9               | 16.87       | 37.1 | 7.96 | 0.21   | 0.674  | 0.49          | 0.29       | —      | —     | -1.41     |
| He 1 bar $Re=500$ (variable properties)     | 500   | 0.159     | 165       | 0.309      | 51.1        | 56.37            | 83.0  | 97.3               | 48.0               | 32.29       | 19.4 | 4.16 | 0.196  | 0.352  | 0.297         | 0.63       | —      | —     | -0.01     |
| He 1 bar $Re=1000$                          | 1000  | 0.318     | 316       | 0.671      | 211         | 50.68            | 76.8  | 92.6               | 47.1               | 28.94       | 21.7 | 4.64 | 0.112  | 0.196  | 0.178         | 0.15       | —      | —     | -0.05     |
| He 1 bar $Re=1500$                          | 1500  | 0.477     | 465       | 1.12       | 521         | 48.79            | 71.7  | 85.9               | 46.8               | 24.77       | 25.3 | 5.42 | 0.0847 | 0.153  | 0.142         | 0.06       | —      | —     | -0.05     |
| He 1 bar $Re=2300$                          | 2300  | 0.731     | 702       | 1.98       | 1.39e+03    | 47.47            | 67.1  | 79.7               | 46.6               | 20.82       | 30.1 | 6.45 | 0.0642 | 0.119  | 0.112         | 0.022      | —      | —     | -0.06     |
| He 1 bar $Re=2300$ (g on)                   | 2300  | 0.731     | 703       | 2.02       | 1.42e+03    | 47.48            | 62.7  | 80.9               | 46.6               | 16.41       | 38.2 | 8.18 | 0.0656 | 0.151  | 0.14          | 0.018      | —      | —     | 0.55      |
| He 1 bar $Re=2300$ (coarse grid)            | 2300  | 0.731     | 702       | 1.95       | 1.37e+03    | 47.47            | 67.1  | 79.9               | 47.3               | 20.85       | 30.1 | 6.44 | 0.0632 | 0.119  | 0.112         | 0.022      | —      | —     | -0.03     |
| He 1 bar $Re=2300$ (fine grid)              | 2300  | 0.731     | 702       | 1.99       | 1.4e+03     | 47.47            | 67.0  | 79.7               | 46.2               | 20.79       | 30.1 | 6.46 | 0.0647 | 0.119  | 0.112         | 0.022      | —      | —     | -0.07     |
| He 1 bar $Re=2300$ (variable properties)    | 2300  | 0.731     | 702       | 1.98       | 1.39e+03    | 47.47            | 66.0  | 76.9               | 46.7               | 19.80       | 31.7 | 6.78 | 0.0643 | 0.125  | 0.117         | 0.022      | —      | —     | -0.01     |
| He 1 bar $Re=5000$                          | 5000  | 1.59      | 1.5e+03   | 6.62       | 9.95e+03    | 46.14            | 55.3  | 61.3               | 46.5               | 9.73        | 64.4 | 13.8 | 0.0461 | 0.117  | 0.11          | 0.0023     | —      | 2.9   | -0.08     |
| He 1 bar $Re=10000$                         | 10000 | 3.18      | 2.99e+03  | 21.8       | 6.51e+04    | 45.57            | 50.3  | 54.3               | 46.4               | 5.00        | 125  | 26.8 | 0.0382 | 0.114  | 0.107         | 0.0003     | —      | 5.6   | -0.18     |
| He 5 bar $Re=2300$                          | 2300  | 0.731     | 140       | 0.395      | 55.5        | 47.47            | 67.1  | 79.8               | 46.6               | 20.82       | 30.1 | 6.45 | 0.0641 | 0.119  | 0.112         | 0.56       | —      | —     | -0.04     |
| He 18 bar $Re=2300$                         | 2300  | 0.731     | 39        | 0.11       | 4.29        | 47.47            | 67.1  | 79.7               | 46.6               | 20.82       | 30.1 | 6.45 | 0.0643 | 0.119  | 0.112         | 7.3        | —      | —     | -0.04     |
| $\text{LN}_2$ 3 bar $Re=500$                | 500   | 4.09      | 5.06      | 0.202      | 1.02        | 78.13            | 85.0  | 89.2               | 77.7               | 7.46        | 84   | 5.77 | 0.163  | 0.151  | 0.14          | 40         | 4.8    | —     | 0.00      |
| $\text{LN}_2$ 3 bar $Re=500$ (g on)         | 500   | 4.09      | 5.06      | 1.68       | 8.5         | 77.32            | 78.9  | 81.7               | 77.7               | 1.74        | 361  | 24.8 | 1.35   | 0.65   | 0.478         | 9.4        | 12.3   | —     | -71.96    |
| $\text{LN}_2$ 3 bar $Re=1000$               | 1000  | 8.17      | 10.1      | 0.501      | 5.07        | 77.56            | 83.1  | 86.6               | 77.6               | 5.77        | 109  | 7.45 | 0.101  | 0.0977 | 0.0931        | 7.8        | 7.4    | —     | 0.00      |
| $\text{LN}_2$ 3 bar $Re=1500$               | 1500  | 12.3      | 15.2      | 0.869      | 13.2        | 77.38            | 82.1  | 85.3               | 77.5               | 4.92        | 127  | 8.75 | 0.0778 | 0.0765 | 0.0736        | 2.9        | 8.7    | —     | 0.00      |
| $\text{LN}_2$ 3 bar $Re=2300$               | 2300  | 18.8      | 23.3      | 1.57       | 36.6        | 77.25            | 81.2  | 84.1               | 77.5               | 4.12        | 152  | 10.4 | 0.0599 | 0.0595 | 0.0578        | 1          | 9.9    | —     | -0.00     |
| $\text{LN}_2$ 3 bar $Re=2300$ (g on)        | 2300  | 18.8      | 23.3      | 3          | 69.8        | 77.22            | 78.5  | 79.9               | 77.5               | 1.35        | 463  | 31.8 | 0.114  | 0.181  | 0.166         | 0.34       | 14.1   | —     | -10.77    |
| $\text{LN}_2$ 3 bar $Re=2300$ (coarse grid) | 2300  | 18.8      | 23.3      | 1.55       | 36.1        | 77.25            | 81.2  | 84.1               | 77.7               | 4.09        | 153  | 10.5 | 0.0592 | 0.06   | 0.0582        | 1          | 9.9    | —     | 0.01      |
| $\text{LN}_2$ 3 bar $Re=2300$ (fine grid)   | 2300  | 18.8      | 23.3      | 1.58       | 36.8        | 77.25            | 81.3  | 84.1               | 77.4               | 4.14        | 151  | 10.4 | 0.0603 | 0.0592 | 0.0575        | 1.1        | 9.9    | —     | -0.02     |
| $\text{LN}_2$ 3 bar $Re=5000$               | 5000  | 40.9      | 50.6      | 5.57       | 282         | 77.11            | 78.6  | 80.0               | 77.4               | 1.55        | 405  | 27.8 | 0.0449 | 0.073  | 0.0704        | 0.083      | 14.0   | 2.9   | 0.04      |
| $\text{LN}_2$ 3 bar $Re=10000$              | 10000 | 81.7      | 101       | 18.6       | 1.88e+03    | 77.06            | 77.8  | 78.6               | 77.4               | 0.81        | 777  | 53.3 | 0.0374 | 0.0699 | 0.0675        | 0.011      | 15.4   | 5.6   | 0.03      |

**Table S8.** Grid convergence index study for both fluids at  $Re = 2300$  (Celik et al. 2008); meshes 86 400 / 291 600 / 984 150 cells, refinement ratio 1.5, production mesh = medium.

| Fluid                   | Quantity                               | Coarse | Medium | Fine   | $p$  | Richardson | $GCI_{\text{medium}}$ | $GCI_{\text{fine}}$ |
|-------------------------|----------------------------------------|--------|--------|--------|------|------------|-----------------------|---------------------|
| He, 1 bar               | $Nu$                                   | 6.439  | 6.449  | 6.457  | 0.64 | 6.485      | 0.70 %                | 0.54 %              |
| He, 1 bar               | $f$                                    | 0.0632 | 0.0642 | 0.0647 | 1.40 | 0.0655     | 2.5 %                 | 1.4 %               |
| He, 1 bar               | $h$ [ $\text{W m}^{-2}\text{K}^{-1}$ ] | 30.05  | 30.10  | 30.14  | 0.64 | 30.27      | 0.70 %                | 0.54 %              |
| LN <sub>2</sub> , 3 bar | $Nu$                                   | 10.513 | 10.436 | 10.382 | 0.87 | 10.26      | 2.2 %                 | 1.5 %               |
| LN <sub>2</sub> , 3 bar | $f$                                    | 0.0592 | 0.0599 | 0.0603 | 1.71 | 0.0607     | 1.6 %                 | 0.8 %               |
| LN <sub>2</sub> , 3 bar | $h$ [ $\text{W m}^{-2}\text{K}^{-1}$ ] | 153.2  | 152.1  | 151.3  | 0.87 | 149.5      | 2.2 %                 | 1.5 %               |

**Property correction and energy accounting.** The helium transport properties of the earlier version of this comparison ( $\mu = 2.73 \mu\text{Pa s}$ ,  $k = 0.0185 \text{ W m}^{-1}\text{K}^{-1}$ ) correspond to helium near 15 K; the present runs use the 50 K values ( $6.36 \mu\text{Pa s}$ ,  $0.0467 \text{ W m}^{-1}\text{K}^{-1}$ , CoolProp/NIST). The mixing-cup outlet temperature is the mass-flux-weighted average  $\sum \phi_f T_f / \sum \phi_f$  over the outlet faces (`surfaceFieldValue`, `weightedAverage` with `phi`); the `weightedAreaAverage` operation, which multiplies the mass flux by the face area a second time, biases the outlet temperature low on a graded mesh and was the cause of the 2–10 % energy imbalances reported earlier. The delivered wall heat flux was verified with the solver-native `wallHeatFlux` function object (9.395 W).

**Wall temperatures and the nitrogen loop pressure.** Wall temperatures are read from the heated-wall patch with `surfaceFieldValue` (`max`, `min`, `areaAverage`; include file `system/wallTstats` in every case), and the wall-profile figure of the manuscript uses the patch face values along the centre row of the first channel (`postProcessing/wallFaces/wall_centreline.csv`). A line sample 0.4 mm inside the fluid, which an earlier post-processing used, underestimates the hottest wall face by 8–10 K for helium and 1–3.5 K for nitrogen and is not used anywhere in the paper. The nitrogen cases are stored under the name `LN2_3bar_*` because their properties were generated at 3 bar; the Boussinesq fields do not depend on the loop pressure (density, viscosity and conductivity of the 77 K liquid change by 0.06, 0.3 and 0.1 % between 3 and 5 bar), and the saturation margins of Table S7 are evaluated at the 5 bar loop pressure of the manuscript, with the 3 bar value (87.9 K) quoted where it matters.

## S3 Hardware efficiencies used in the real ledger

**Table S9.** Stage efficiencies and the nitrogen supply-energy multiplier used in the real ledger.

| Stage                                     | $\eta$ range                                     | Source                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 K cryocooler stage                      | 0.008–0.015                                      | PT415: 1.5 W at 4.2 K with 40 W at 45 K for 10.7 kW [2]: the two stages together return 0.031 of their combined Carnot work, which the per-stage ranges reproduce to within 7 % at their midpoints; Radebaugh: “at 4 K efficiencies of about 1 % of Carnot or less are typical” [3]                        |
| He first stage (45–80 K)                  | 0.10–0.15                                        | Radebaugh: 10–15 % of Carnot for the best commercial 80 K coolers; large plants up to 30 % [3, 4]                                                                                                                                                                                                          |
| LN <sub>2</sub> supply (basis of Table 6) | $M_{N_2} = 9.1\text{--}12.7$ W per W intercepted | plant consumption 0.5–0.7 kWh kg <sup>−1</sup> [5] divided by the latent heat 199 kJ kg <sup>−1</sup> at 1 bar; equivalent to 0.30–0.40 of Carnot at 77 K (ideal liquefaction work 0.213 kWh kg <sup>−1</sup> [6]); the fraction-of-Carnot form gives 7.2–9.7 W/W and changes the nitrogen totals by 1–2 % |

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